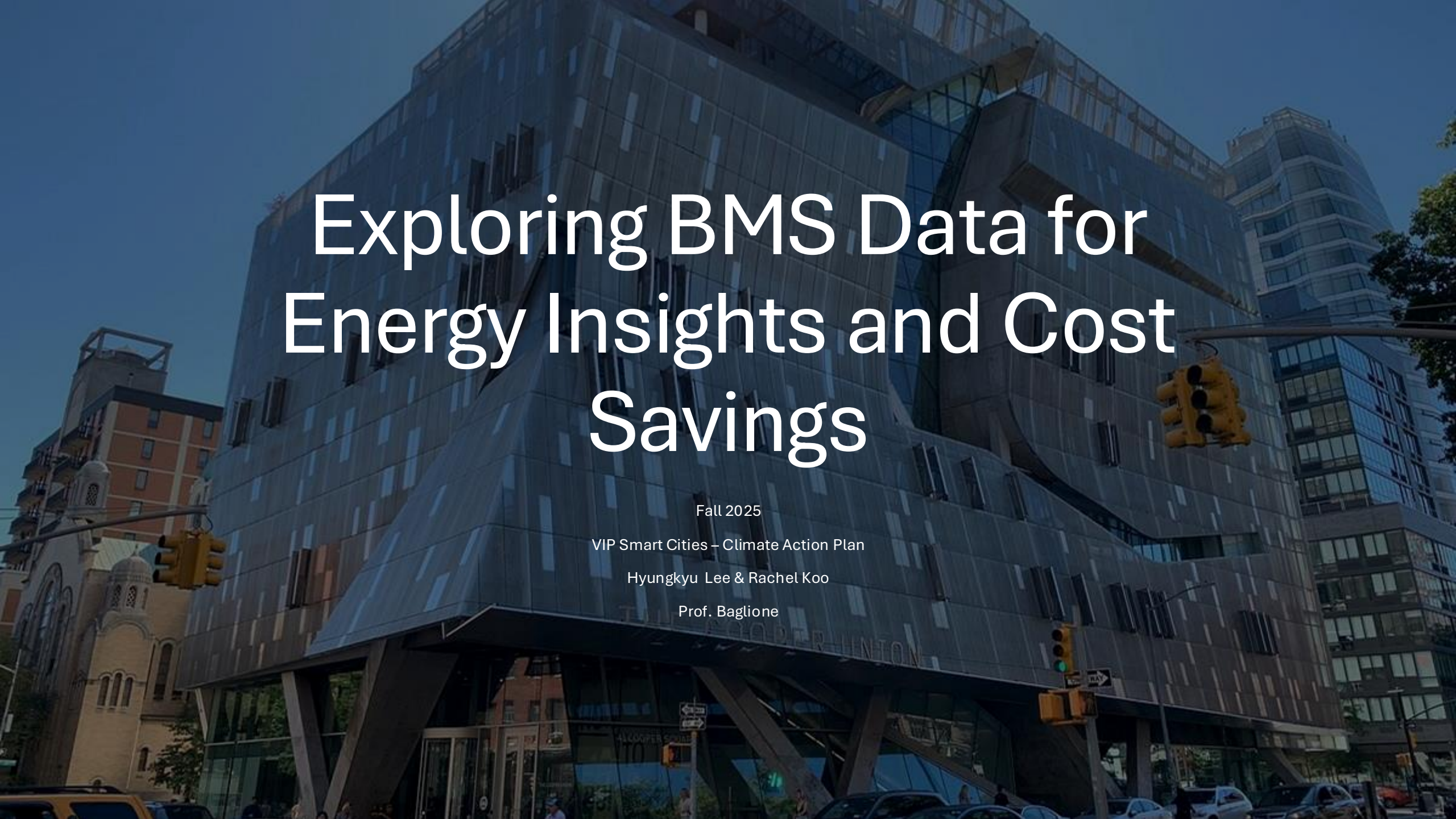


Exploring BMS Data for Energy Insights and Cost Savings

Fall 2025

VIP Smart Cities – Climate Action Plan

Hyungkyu Lee & Rachel Koo

Prof. Baglione

41 Cooper Sq Carbon Emission (Lab Carve Out)

nyc LL97 carbon emissions calculator

41 Cooper Square

manual user input

Building Inputs

	Building Input	Area (SF)	
1	College/Univ...	134,142	x
2	Laboratory	42,470	x

+ Add Building Type

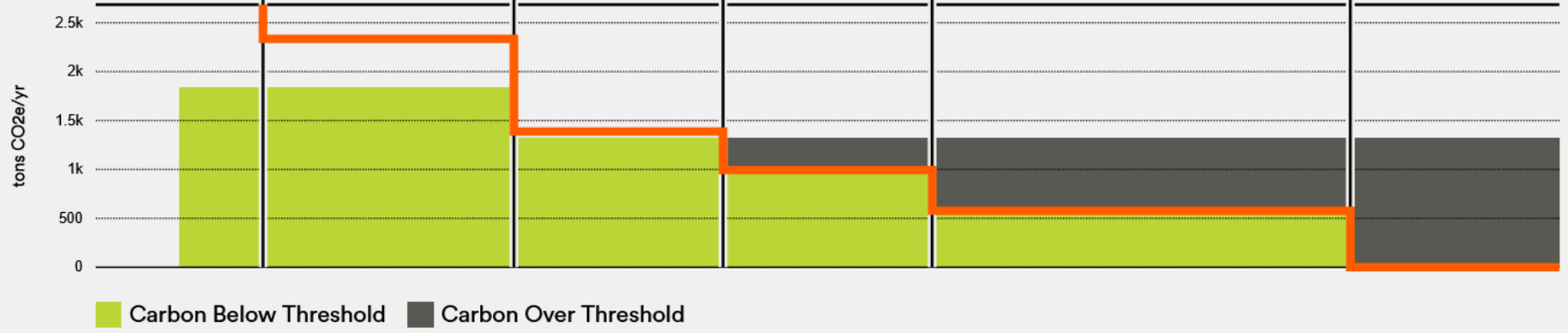
Utility Inputs

Electricity (kWh)	$/kWh
3,589,224	0.22
Natural Gas (therms)	**$/therm**
150,928	0.997
Steam (mLbs)	**$/mlb**
0	35
Fuel Oil #2 (gal)	**$/gal**

carbon | cost

Annual Carbon Threshold Summary

	2024-2029	2030-2034	2035-2039	2040-2049	2050-
Emissions (tCO2e/yr)	1,839	1,322	1,322	1,322	1,322
Threshold (tCO2e/yr)	2,335	1,387	995	577	0
Est Penalty ($/yr)	$0	$0	$87,661	$199,674	$354,310



units: absolute | normalized group: summary | end uses

Calculator engine by AKF Group LLC

41 Cooper Sq Carbon Emission (w/o Lab Carve Out)

nyc LL97 carbon emissions calculator

41 Cooper Square

manual user input

Building Inputs

	Building Input	Area (SF)
1	College/Univ...	176,672

Add Building Type

Utility Inputs

Electricity (kWh)	$/kWh
3,589,224	0.22
Natural Gas (therms)	$/therm
150,928	0.997
Steam (mLbs)	$/mlb
0	35
Fuel Oil #2 (gal)	$/gal
0	1.65
Fuel Oil #4 (gal)	$/gal
0	1.65

Use Default Rates

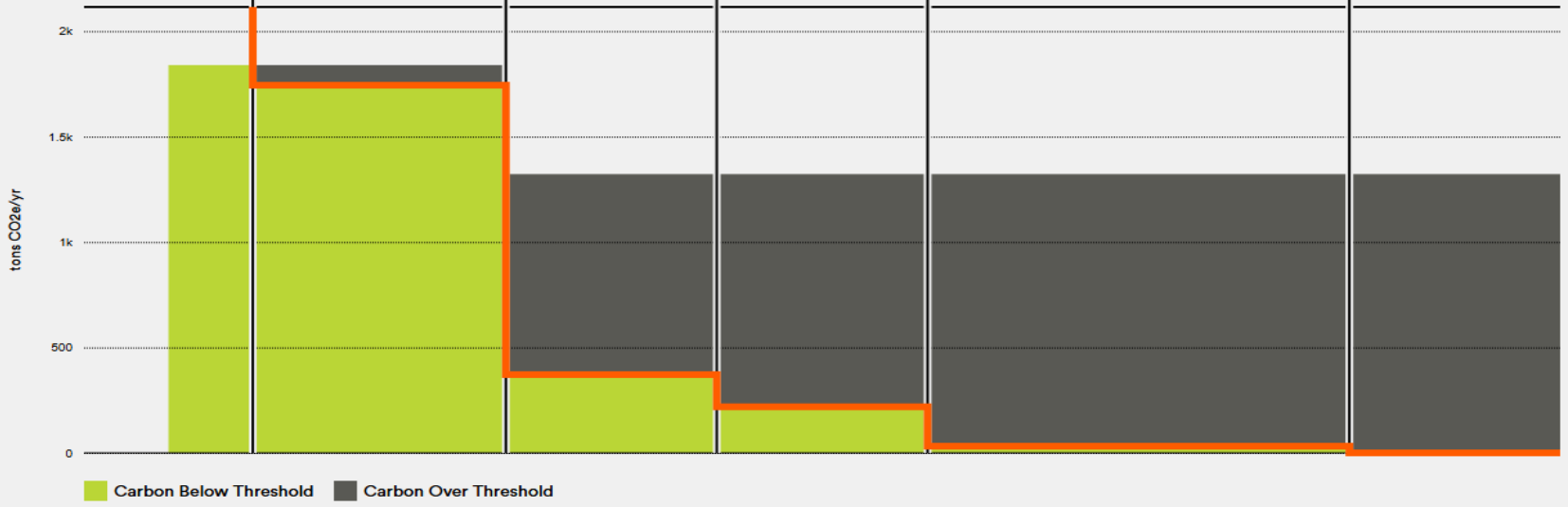
Carbon Deductions

Solar PV (kWh) 0

carbon | cost

Annual Carbon Threshold Summary

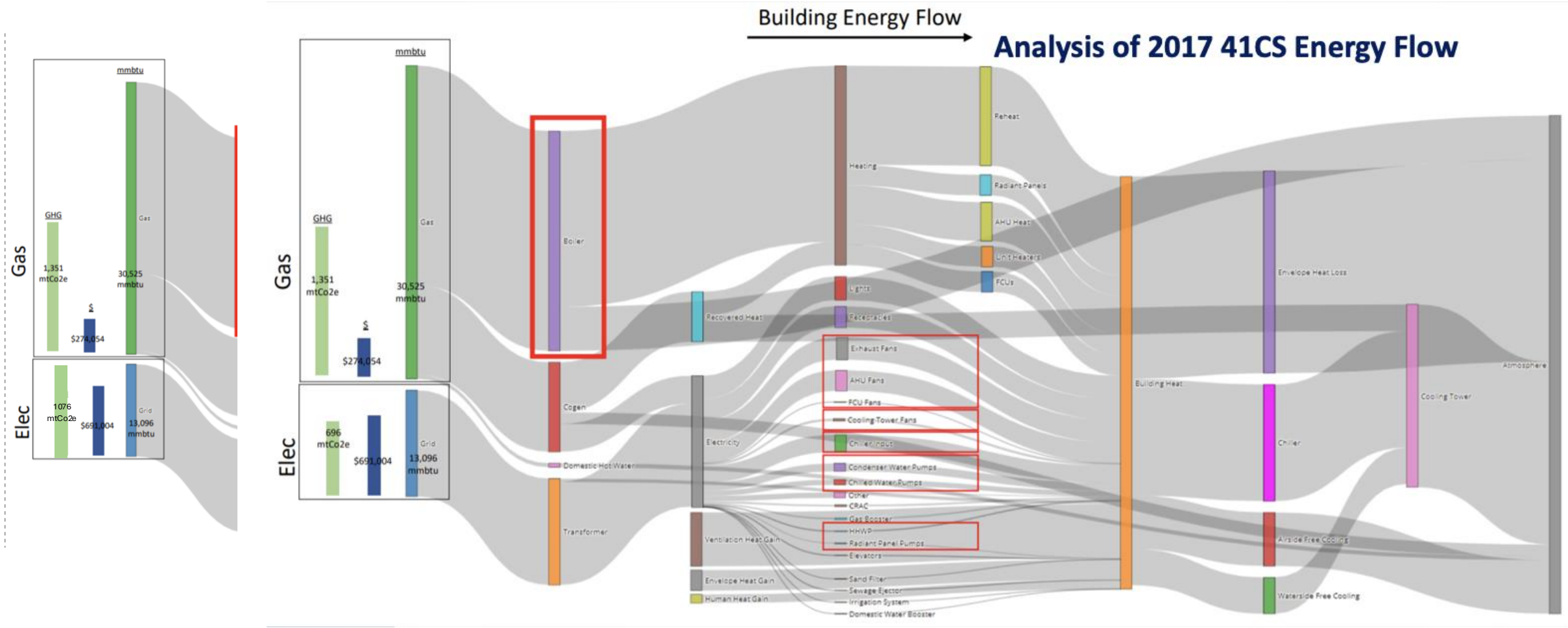
	2024-2029	2030-2034	2035-2039	2040-2049	2050-
Emissions (tCO2e/yr)	1,839	1,322	1,322	1,322	1,322
Threshold (tCO2e/yr)	1,744	371	218	32	0
Est Penalty ($/yr)	$25,454	$254,891	$295,772	$345,749	$354,310



units: absolute | normalized group: summary | end uses

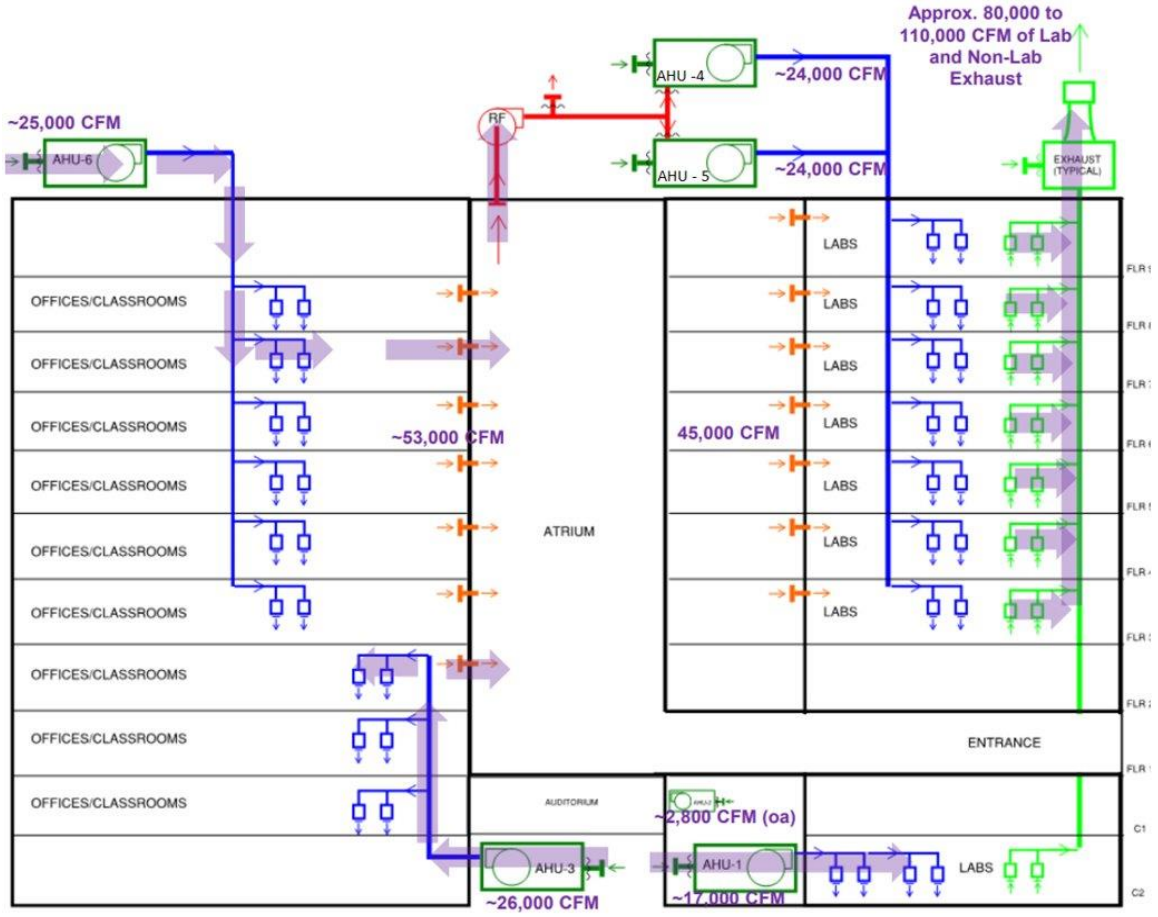
Calculator engine by AKF Group LLC

Sankey 2017 w/ Updated Carbon Coefficient

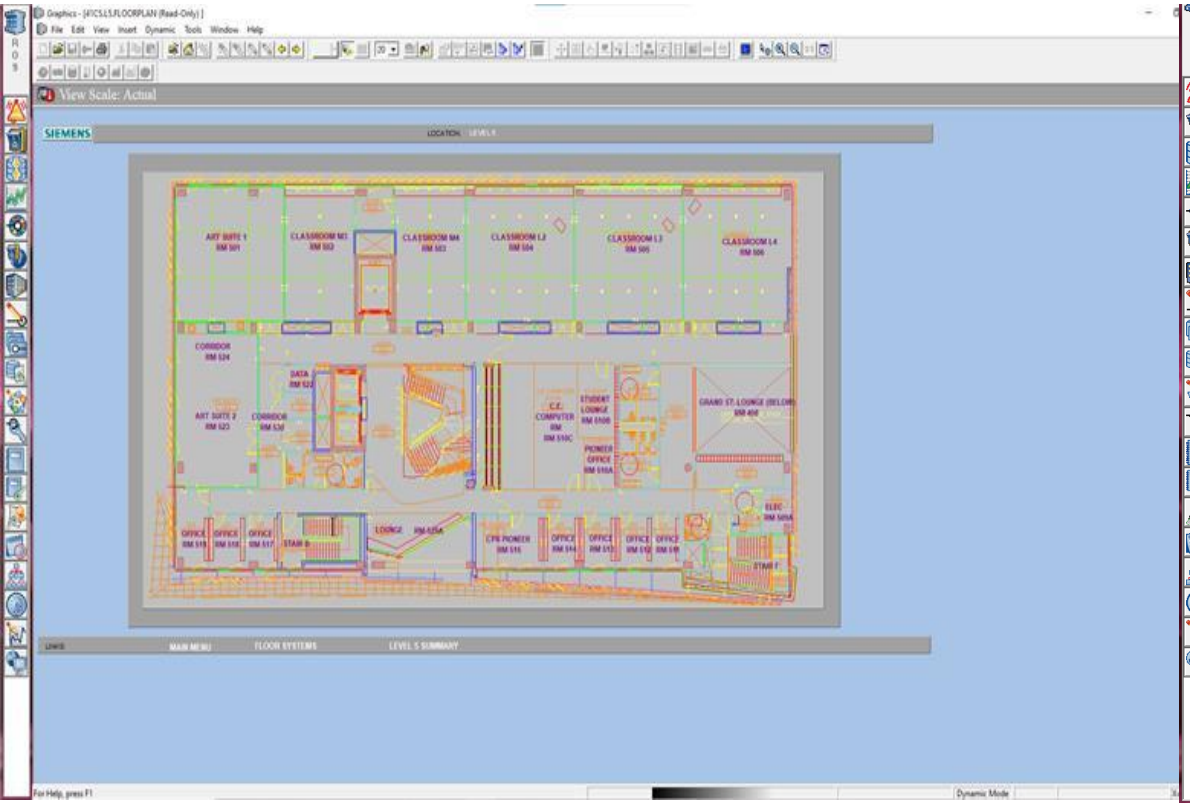


Air Handling Unit (AHU) Serving Areas

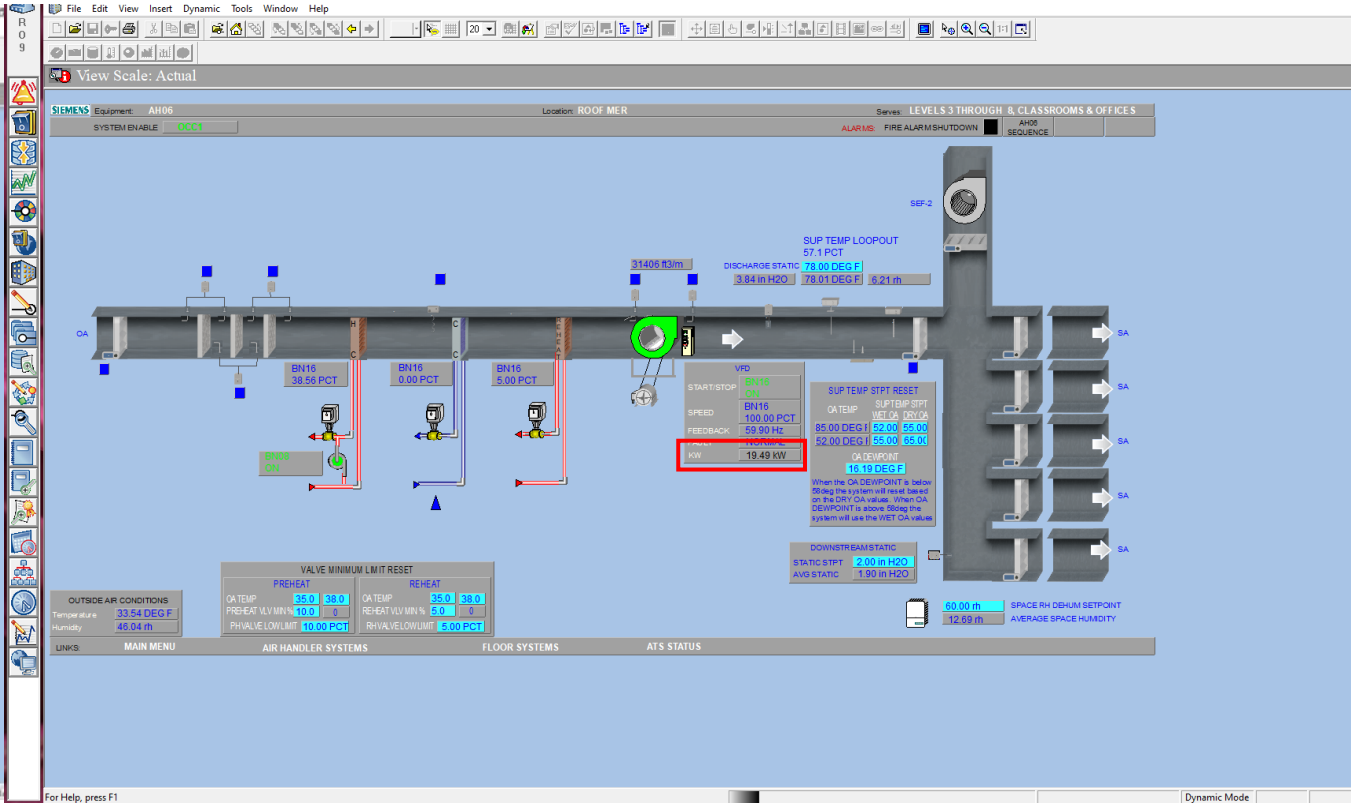
- AHU 1 outside → lower labs
- AHU 2 → rose auditorium
- AHU 3 outside → 1, 2, 3 floor
- AHU 4 atrium → labs
- AHU 5 atrium → labs
- AHU 6 outside → 3 and above



Building Management System (BMS)



Floorplan of 5th



AHU 06

Air Handling Units (AHU)

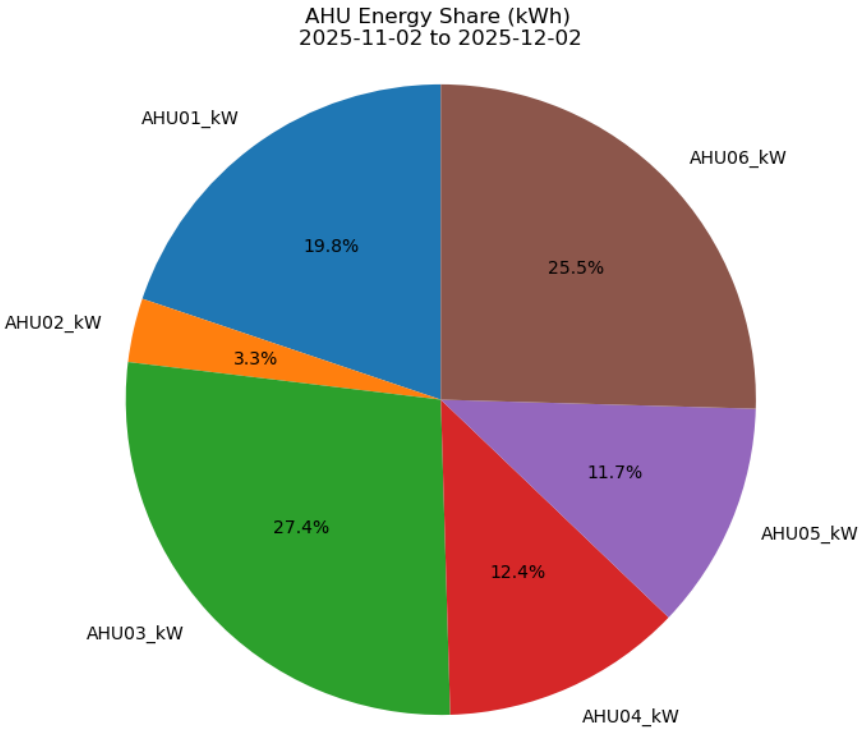
In commercial buildings HVAC systems use ~40 %[1)] of total building energy use.



- AHU kW data of AHU01 – AHU 06 from 2025-10-05 to 2025-12-05 from BMS.
- Took the average kW values that fall inside each 15 minute bin.

1) WBDG – High-Performance HVAC, National Institute of Building Sciences, https://www.wbdg.org/resources/high-performance-hvac

AHU Energy Share



AHU	kWh
AHU01	11,139.51
AHU02	1,844.96
AHU03	15,394.73
AHU04	6,962.84
AHU05	6,565.76
AHU06	14,312.72
Total	56,220.53

- Total AHU energy use (Nov 2 – Dec 2, 2025) : 56,220.53 kWh
- Largest energy consumers:
  - AHU03 (27.4%) 1, 2, 3 floor
  - AHU06(25.5%) floor 3 and above
  - AHU01 (19.8%) lower floor

AHU Grid Energy Share

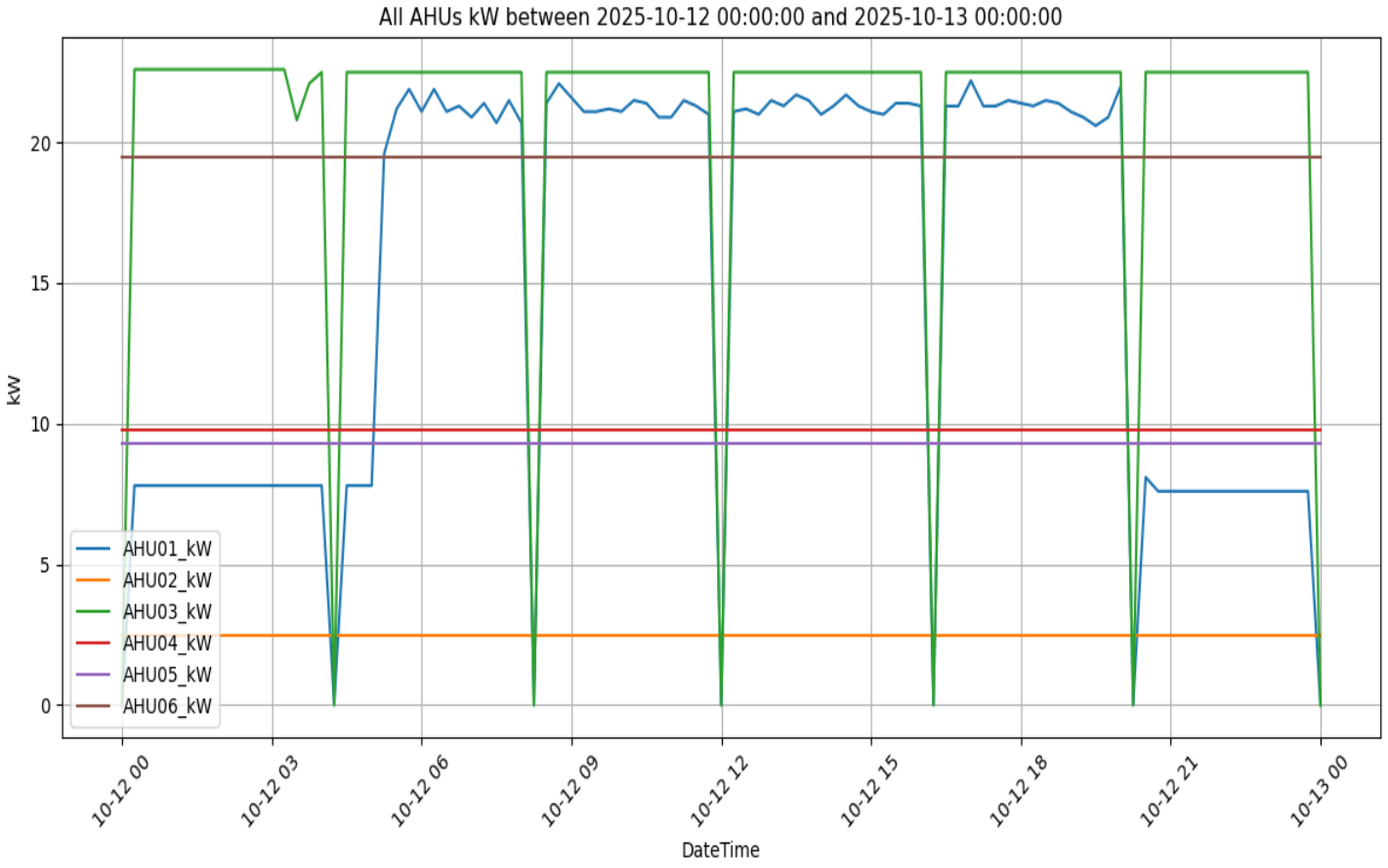
Source	Month - Year	Electric - Grid (kwh)	AHU %
ESPN	Nov-22	237,622.99	23.66%
	Nov-23	255,936.62	21.97%
	Nov-24	274,329.05	20.49%
Bills	Nov-22	268,000.00	20.98%
	Nov-23	255,936.71	21.97%
	Nov-24	274,329.14	20.49%

•The building's total electricity consumption for Nov 2025 is not yet available from billing or ESPN datasets.

• Used prior years (2022–2024) as reference baselines to estimate AHU share of building load

•AHUs make up ~20–24% of total building electricity.

24 hr AHU Power Consumption trend



Trends

- AHU03 and AHU06 operate at highest power levels with periodic shutdown cycles.
- AHU01's energy load decreases during the nighttime as HVAC demand changes.
- AHU02, AHU04, and AHU05 remain very stable.

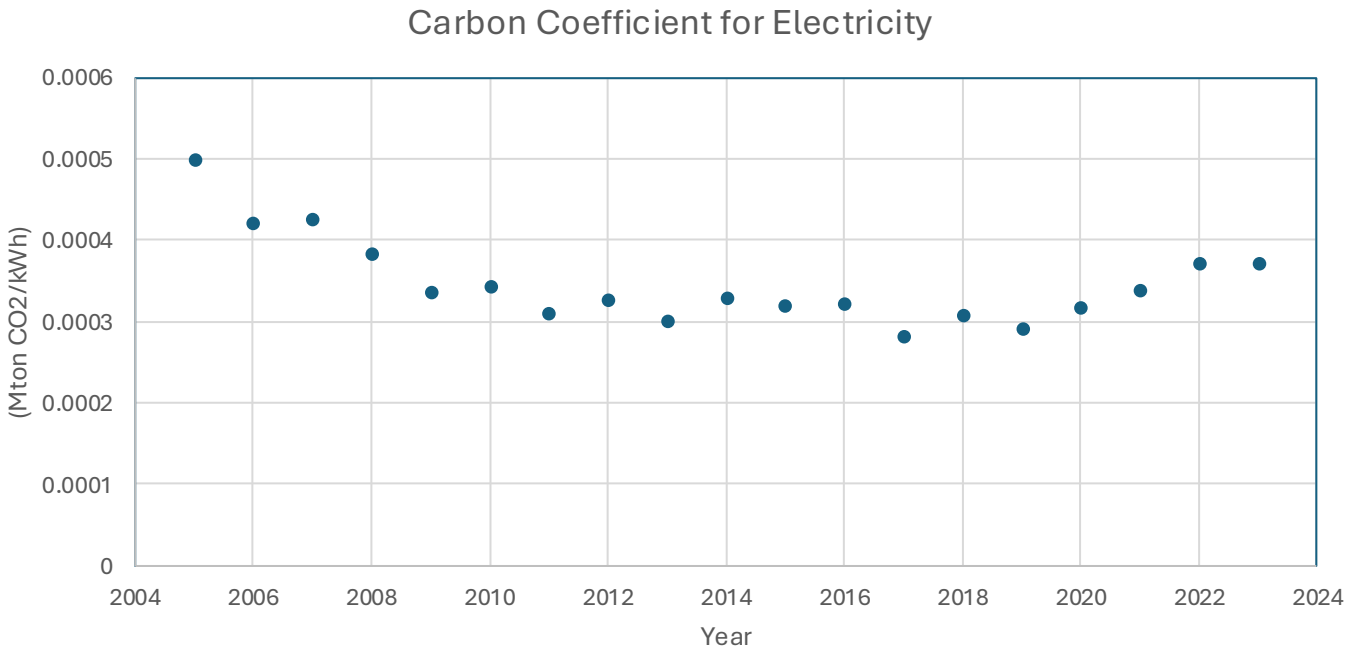
Opportunities for Energy Savings

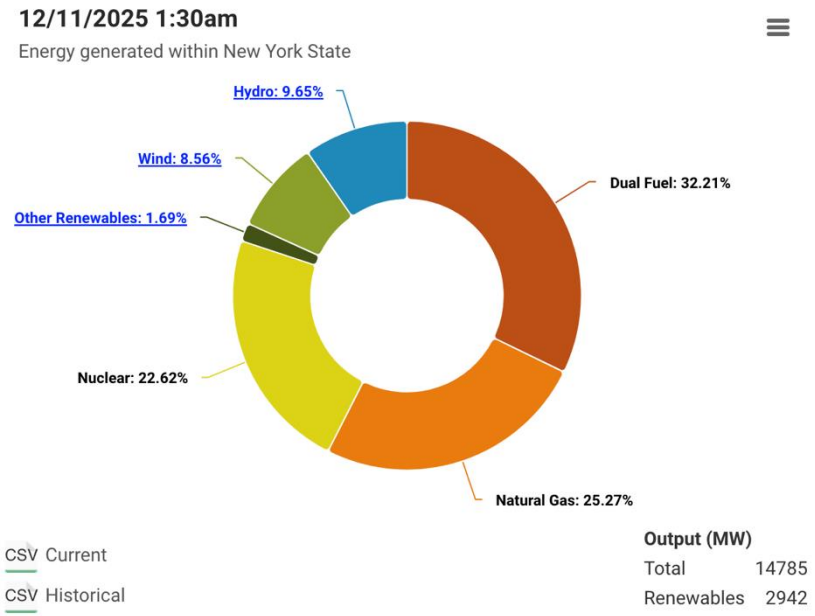
- AHU02 supplies outside air to the rose auditorium.
  - ➢ The AHU02 is always on
    - Occupancy based HVAC system.
    - Reservation based

Emission Comparison

Trends in Carbon Coefficient for Electricity

1)



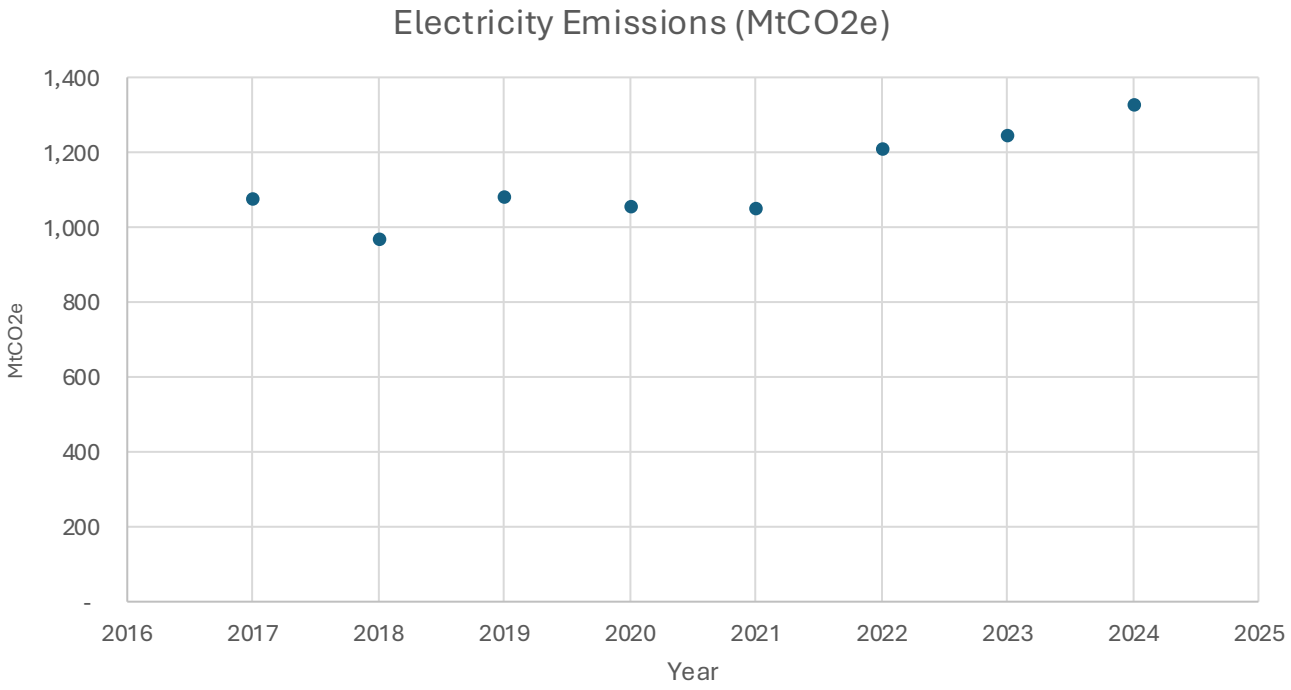


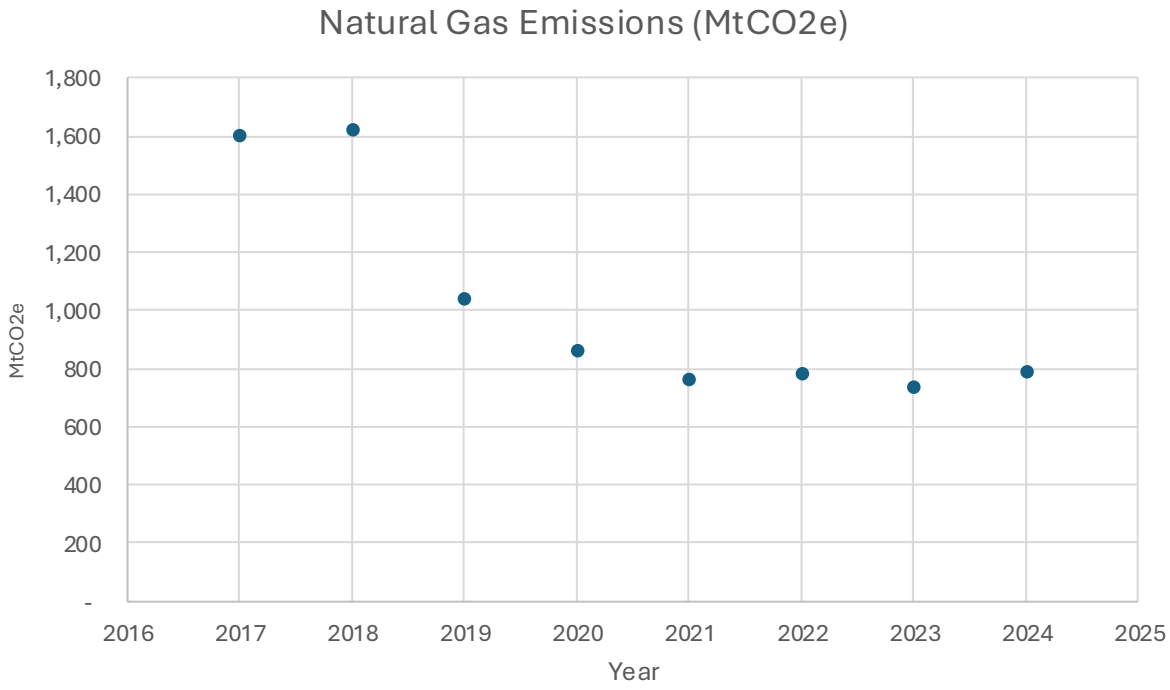
Decrease in carbon coefficient factor. But increase after 2020 due to closing Indian Point Nuclear Plant.

1) Data source: NYISO Real-Time Dashboard, https://www.nyiso.com/real-time-dashboard

Emission Comparison

Carbon Emissions calculated using NYC's (proposed) emission factors for 2024





- Natural gas emissions dropped sharply between 2018 and 2019 following cogeneration retirement.
- Electricity-related emissions increased after 2021 as carbon emission factors rose.

Takeaways

Key Learnings:

- Energy consumption is highly uneven across AHUs, with AHU03 and AHU06 dominating total load due to the zones they serve
- Time-series BMS data reveals operational patterns (night setbacks, shutdown cycles, always-on behavior) that are not visible from monthly utility bills alone
- As NYC grid emissions fluctuate, electrification alone is not sufficient
    - operational efficiency remains critical.

Next steps:

- Expand analysis to seasonal behavior (cooling vs heating months)
- Integrate weather and occupancy data to explain load drivers
- Develop a prioritized retrofit or controls roadmap for high-impact AHUs


Thank you